Does Grounding Decay with Distance? Measuring and Repairing Weak Links in LLM Causal Explanations of Indian Macroeconomic News

Project Proposal, Language Models & Agents (LMA) Group Project 2026

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1 Project Title

Does Grounding Decay with Distance? Measuring and Repairing Weak Links in LLM Causal Explanations of Indian Macroeconomic News. We study whether evidential support weakens and causal language strengthens as an LLM builds longer causal explanations, and whether an agent that verifies each link and searches for evidence on weak links can repair this.

2 Domain and Target User

The domain is Indian macro-financial news on crude oil, the rupee, RBI policy and bond yields, using a frozen corpus from 1 March to 30 September 2026. The target user is a student or non-expert reader who wants to know what is connected to what, why, and how far the evidence supports each link. The system gives no investment advice and makes no forecasts of its own.

3 Problem Statement

When an LLM explains how economic events are connected, links far from the original event may rest on less evidence while being stated just as confidently, and readers cannot see where the explanation stops being grounded. Readers increasingly ask LLM assistants to connect events reported separately in the news, such as an oil shock, a weaker rupee and an RBI policy decision. Early links may be directly supported by retrieved evidence, while later links may increasingly rely on unsupported model reasoning, and hedged source language may harden into firm causal claims along the chain. Whether this degradation actually increases with hop distance, and whether targeted retrieval can reduce it, is what we test. We study this by representing each explanation as an event-impact graph in which every edge is labelled source-asserted or model-inferred and given a strength, which lets us measure support hop by hop (Figure 1).

4 Research Questions

RQ1 (phenomenon): In LLM-generated explanations of Indian macro-financial events, do unsupported causal links and strength inflation increase with hop distance from the queried event?

RQ2 (intervention): Can the full graph-and-repair system reduce unsupported links and strength inflation, especially at deeper hops, while keeping useful recall, compared with standard RAG and with RAG prompted to label its own links under the same model, first-stage retriever, corpus and date boundary?

Hypotheses:

H1: In standard RAG (B1), the unsupported established-edge rate (UER, Section 12) and strength inflation increase with hop distance; the same trend is reported for prompted labelling (B2) as an exploratory comparison.

H2a: M has at least 10 percentage points lower overall UER than B1.

H2b: The increase in UER and strength inflation from shallow to deeper hops is smaller for M than for B1.

H2c: M has lower UER and strength inflation than B2.

H2d (recall guard): M's reference-edge recall is no more than 10 points below B1's.

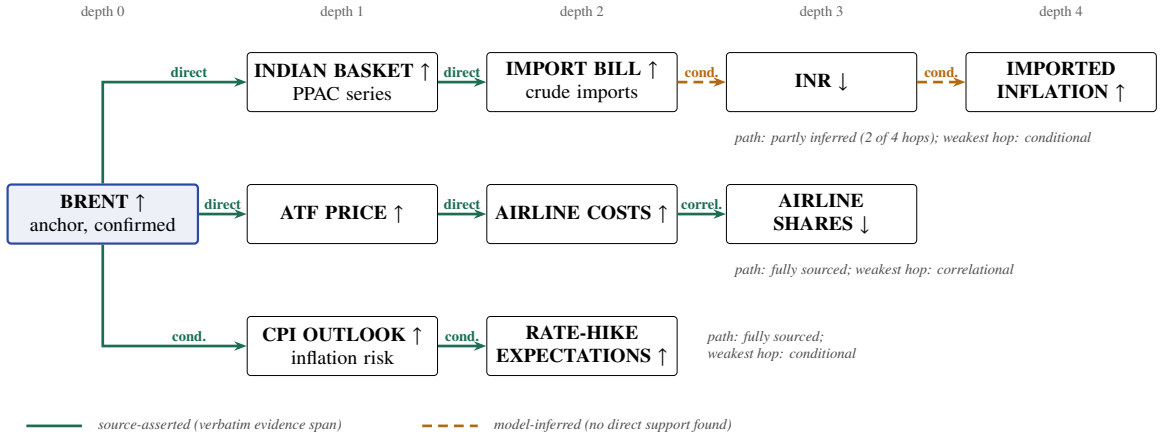


Figure 1: Illustrative example, for "Brent crossed $100 on 12 March 2026; what effects on India were sources discussing?". A fluent RAG answer presents all three chains with equal confidence. Our graph separates source-stated edges (solid) from model reasoning (dashed): the airline chain is fully sourced, but two hops of the rupee chain have no direct support.

5 Why the Problem Matters in This Domain

Macroeconomic reporting frequently uses causal and quasi-causal explanations. Readers do not need another summary of these events, they need to see how the events are linked and how much of each link the evidence actually supports. We build the event-impact graph for this reason. It turns an explanation into separate, inspectable links, so that a student can see which links a source states, which the model added from its own reasoning, and how strongly each link is claimed. A reader can then follow a chain from cause to effect, open any link to read its evidence, and see where a fluent explanation rests on an unsourced step. In practice, the graph helps readers separate what was reported from what was assumed.

This matters in India today. India imports most of its crude oil, and in March 2026 reports stated that about 90% of its LPG imports pass through the Strait of Hormuz (Takshashila Institution, 2026). Reports linked the 2026 oil shock to rupee weakness, with a record low near INR 96.9 per USD (Republic World, 2026), and to inflation risks, fuel prices and RBI policy expectations. Indian market reports often connect such events with "amid" or "after", which suggests causation without stating it, and RBI statements are deliberately conditional. When an assistant turns this language into firm causal chains, readers come away with explanations that sound sourced but are not, and the graph is designed to make that gap visible.

6 What Is Missing in Existing Approaches

Causal graphs have already been built from news. Causal Cartographer (Gendron et al., 2025) extracts a 975-node causal graph from oil-price news, but its extraction prompt asks for relationships "based on the text and common sense knowledge" and does not record which edges came from the text. WorldReasoner (Chi et al., 2026) attaches article-level evidence to event graphs but does not check evidence edge by edge, and CausalRAG (Wang et al., 2025) compares graph retrieval with RAG using LLM-judged faithfulness. Sentence-level support labels exist (Asai et al., 2024; Niu et al., 2024; Gao et al., 2023b), as does per-claim iterative verification (Xie et al., 2025; Wei et al., 2024; Gao et al., 2023a). The direct, conditional and correlational labels come from Yu et al. (2019), and ReCEval (Prasad et al., 2023) scores a reasoning chain by its weakest step. SemEval-2026 Task 12 (SemEval-2026 Task 12 organisers, 2026) evaluates direct-cause selection from news but does not label or verify individual links.

No prior work we found measures how evidential support and causal strength degrade hop by hop in LLM-generated causal explanations of real news, or tests whether targeted verification and retrieval repair that

degradation. Provenance labels (source-asserted or model-inferred), set only after passage-level verification, are the mechanism we use to measure it. We do not claim novelty for causal graph extraction or for iterative search.

7 Capabilities Used and Why the Problem Needs Each

Table 1 summarises the capabilities; the metrics for each are given in Table 4. Fine-tuning is not used, because the research questions concern how evidential support changes along an explanation and whether verification and targeted retrieval repair it, not model training.

Table 1: Capabilities and why each is needed; Table 4 gives the metrics.

Capability	Why needed	Implementation
RAG	Source-backed claims must be grounded in the frozen 2026 corpus, so relevant evidence must be retrieved	Frozen corpus with an as-of date filter; BM25 or hybrid retrieval *[needs pilot]*; cross-encoder reranker tested in baseline B1-R
Reasoning	Decide whether a passage states A→B, its direction and strength; compose paths	Rule checks plus an LLM verifier; weakest-link path labels
Agentic	Decide which edge to investigate and when to stop within a budget	Single agent with explicit state and a priority policy
Tool calling	Search, data lookup, verification and graph edits must be correct and safe	Six read-only, schema-validated tools

8 Proposed Idea and Architecture

Figure 2 shows the proposed system. Given a query, the agent identifies the anchor event and an as-of date, confirms the move from a macro series, retrieves passages, and extracts events and candidate edges into a closed list of about 40 variables (for example `BRENT`, `INR_USD`, `CPI_HEADLINE`) with a direction (up, down or stable). It then proposes bridging links that it believes are missing; these start as model-inferred.

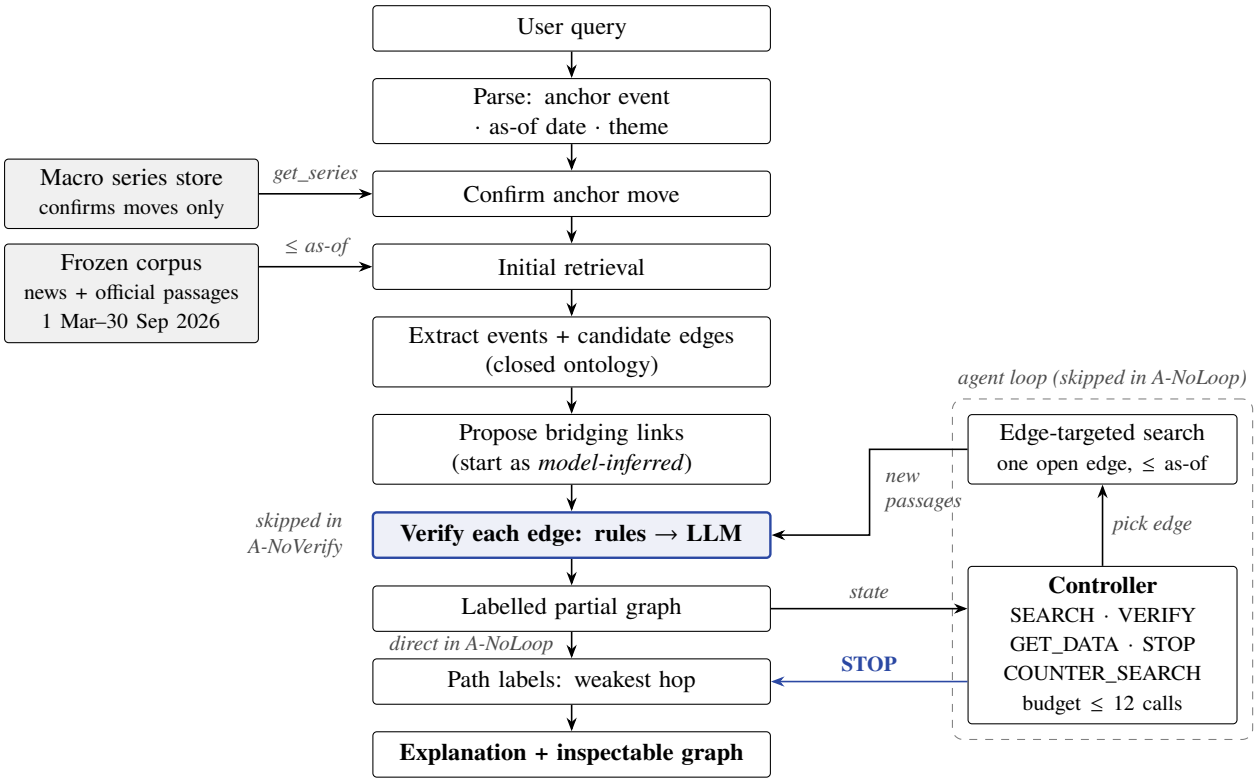


Figure 2: Proposed system architecture. The corpus is accessed only through the as-of filter, and macro series only confirm that a move occurred. In M and A-NoLoop, only the verifier (highlighted) can mark an edge source-asserted. The dashed region is the agent loop. Removing the verifier gives ablation A-NoVerify, removing the loop gives A-NoLoop.

Edge labels: Each edge carries the labels in Table 2 and, when asserted, a verbatim evidence span with its source. Three coarse strength levels are used because graded causal strength shows low annotator agreement (Romanou et al., 2023).

Table 2: Edge labels, adapted from Yu et al. (2019) and the Causal News Corpus guidelines (Tan et al., 2022). For scoring, strength is ordered Direct > Conditional > Correlational.

Label	Meaning (typical cues)
Source-asserted	A passage published by the as-of date states the relation in this direction; set only by the verifier
Model-inferred	No supporting passage was found and verified in the permitted corpus under the search procedure; the link is the model's reasoning (unsourced, not necessarily wrong)
Direct	Unhedged causal language (caused, due to, driven by, weighed on)
Conditional	Hedged or conditional causal language (could, may, if, upside risk), or the cause itself is uncertain
Correlational	Co-occurrence or sequence only (amid, as, after, alongside)

Verification and paths: The verifier first applies rules: the passage must be in the corpus and dated on or before the as-of date, must mention both variables, and must contain the quoted span word for word. The same LLM, in a fresh context and without the generator's reasoning, then judges whether the passage states the relation and at what strength. A path takes the strength of its weakest hop, as in ReCEval's minimum aggregation (Prasad et al., 2023), and is shown as fully sourced only if every hop is source-asserted (Figure 1).

Agent loop: The controller keeps a state of open edges, search counts and remaining budget. It verifies unverified edges first, then searches specifically for evidence on the most important model-inferred edge, adapting the retrieve-and-verify loop of FIRE (Xie et al., 2025) to graph edges. For "why" questions it also looks for a second, competing cause. It stops when the important edges are resolved, when two searches add nothing new, or after 12 tool calls. The tools are `search_corpus`, `get_series`, `verify_edge`, `counter_search`, `update_graph` and `stop`. All are read-only, and the date filter and budget are enforced in code.

Model and retrieval: One model is used for every condition. The prior choice is Qwen3-8B (Qwen Team, 2025) at 4-bit precision, with Qwen3.5-9B (Qwen Team, 2026) as the alternative and Claude Haiku 4.5 as an API fallback *[needs TA decision]*. A week-1 pilot on 20 examples per task (extraction, strength, JSON validity, tool choice and rejecting unsupported edges) decides the model *[needs pilot]*. A second 20-pair pilot chooses between BM25, BM25 with alias expansion, bge-m3 (Chen et al., 2024) and a hybrid with reciprocal rank fusion (Cormack et al., 2009), taking the simplest option within 0.03 Recall@10 of the best *[needs pilot]*. A cross-encoder reranker (BAAI, 2024) is then evaluated on top of the chosen retriever, both on retrieval metrics and end to end as baseline B1-R.

The prototype will implement this pipeline as a read-only agent over the frozen corpus, with an interactive event-impact graph as its final output.

9 Expected Novelty

- **Hop-by-hop measurement:** Measure how evidential support and causal strength change across successive hops in LLM-generated causal explanations of real news.
- **Verification and retrieval evaluation:** Test whether edge verification and edge-targeted retrieval reduce the degradation in evidential support and causal strength across hops.
- **Edge schema and annotation guide:** Develop an edge-level schema containing provenance and causal-strength labels, together with an annotation guide tailored to Indian macroeconomic news.
- **Evaluation dataset:** Create a small evaluation set of 30 anchored 2026 queries with reviewer-identified evidence graphs.

We measure whether textual evidence supports a causal claim; we do not establish whether the relationship is a true economic causal effect.

10 Dataset and Evaluation Environment

Corpus: About 900–1,200 de-duplicated articles from BusinessLine, Economic Times, Business Standard and Mint, plus 60–90 official documents (RBI policy statements and minutes, MOSPI CPI releases, PPAC and PIB releases), published between 1 March and 30 September 2026. Macro series (Brent, Indian crude basket, USD/INR, CPI, bond yields, FPI flows) only confirm that a move happened, they are never evidence for an edge.

Custom evaluation set: Queries are anchored on eight episodes, including the March oil shock, the April ATF price increase, the May rupee low, the June and August RBI meetings and the September bond sell-off. These episodes were found in news reports and will be confirmed against official data *[needs pilot]*. There are 6 dev and 24 test queries, half asking why something happened and half asking what effects were discussed, each with an as-of date. For each query we build a reviewer-identified evidence graph of about 4–6 source-asserted edges with evidence spans. We also build a 125-item challenge set of 100 corrupted edges (reversed direction, inflated strength, unsupported intermediate hop, wrong date, wrong target) and 25 clean controls, 40 frozen agent states with gold next actions, 20 scripted tool faults (empty results and errors), and 25 safety prompts. Two team members annotate independently on a shared subset to measure agreement (Cohen's κ), using the five causality tests of the Causal News Corpus (Tan et al., 2022).

Public datasets: None is used for training. FinCausal (Mariko et al., 2020) and the causal-language data of Yu et al. (2019) provide 200-item checks of causal detection, direction and strength. IndiaFinBench (Pall, 2026) is a domain reference only, since it does not test causal propagation.

11 Baselines

All conditions share the model, corpus, date filter, first-stage retriever, queries and prompt-tuning budget (Table 3). B1-R differs only by applying a cross-encoder reranker before selecting the final top-k passages. It is included to test whether improved evidence ranking alone can close the gap with M; it is not part of the primary M-versus-B1 hypothesis. The prose answers of B1 and B1-R are converted to edges by LLM pre-extraction with full human correction; a claim counts as established if it is cited or stated without hedging. Support is judged blind, on shuffled edges with the condition hidden.

Table 3: Experimental conditions.

Condition	Description	Role
B0 Closed-book	Query only, no retrieval	Reference only
B1 Standard RAG	Top-k passages, one call, prose with citations	Main baseline
B1-R RAG + reranker	Same first-stage retriever; top-50 candidates re-scored by a cross-encoder reranker (BAAI, 2024) and cut to the same top-k; one call, prose with citations	Reranker ablation / stronger RAG baseline
B2 RAG + self-labels	Same context as B1, one call, outputs edges with its own provenance and strength labels	Prompting control
A-NoVerify	Full system with the verifier replaced by self-labels	Verifier ablation
A-NoLoop	Full system that retrieves once (no agent loop)	Agentic ablation
M Full system	Graph, labels, verifier, agent loop and weakest-link paths	Proposed method

12 Metrics, Including Capability-Specific and Safety Metrics

Primary metric: For each test query q, let E_q be the edges a system presents as established (edges shown as source-asserted for M, A-NoVerify and A-NoLoop; edges labelled as sourced for B2; cited or unhedged claims for B1 and B1-R). Then

$$\mathrm{UER}_q = \frac{\left|\{e \in E_q : e \text{ is not supported by its attached evidence}\}\right|}{|E_q|}, \tag{1}$$

averaged over test queries. An edge is supported if its cited passage states the relation in the same direction and was published by the as-of date.

Secondary metrics: Reference-edge recall (which also guards against abstention), strength inflation rate (among established edges that are supported, the fraction whose presented strength exceeds the strength stated in the evidence; unsupported edges are counted by UER), provenance accuracy against human labels, and strict path validity. All four are also reported by hop distance from the anchor event (1, 2, and 3 or more).

Analysis by research question: For RQ1, UER and strength inflation are reported at hop 1, hop 2 and hop 3 or deeper. For RQ2, M is compared with B1 on overall UER, overall strength inflation, deep-hop degradation and reference-edge recall; with B1-R to test whether better ranking alone closes the gap; with B2 on UER and strength inflation; with A-NoVerify to test whether verification helps; and with A-NoLoop to test whether targeted retrieval helps.

Capability-specific metrics: UER, strength inflation and recall answer the research questions end to end. Table 4 separates the metrics that test each capability on its own, and the comparison that isolates its effect on the end-to-end results. We also report qualitative examples and a failure analysis.

Table 4: What each capability is tested on, with which metrics, and which comparison isolates it.

Capability	What is tested	Metrics (test data)	Isolating comparison
RAG	Does retrieval surface and rank the reviewer-identified evidence passage for each link?	Recall@5, Recall@10 and MRR, with and without the reranker (gold evidence of the test reference edges)	Retriever vs retriever + reranker; B1 vs B1-R end to end
Reasoning	Does the verifier judge support and strength correctly, and are paths labelled by their weakest hop?	Corruption detection rate and false-alarm rate on clean controls (125-item challenge set); strength macro-F1 (human-labelled edges and Yu et al. (2019) checks); path-label accuracy (fixed labelled paths); direction and provenance accuracy and FinCausal checks as diagnostics	M vs A-NoVerify
Agentic	Does the controller choose the right next step and stop sensibly?	Next-action accuracy (40 frozen states with gold actions); redundant-search rate, unresolved high-priority edges at stop, tool calls per query (test-run logs)	M vs A-NoLoop
Tool calling	Are calls well formed, valid and safe?	Tool-selection accuracy, argument validity (schema and as-of date), recovery from empty results and errors, unsafe-call rate (frozen states plus 20 scripted faults)	Measured directly
Safety	Does the system stay read-only and avoid advice and forecasts?	Advice leakage, own-forecast leakage, post-date evidence use, asserted labels without a verbatim span (25 adversarial prompts)	Measured directly

Statistics: Human labels cover B1, B2 and M; an LLM judge scores B1-R and the ablations only if it agrees with humans at $\kappa \geq 0.6$ *[needs TA decision]*. UER is the primary outcome and strength inflation is a pre-specified secondary outcome. Two tests are pre-declared on UER: for RQ1, the difference between hop 3 or deeper and hop 1 in B1; for RQ2, the difference in overall UER between M and B1. Both use a paired bootstrap over queries with 95% confidence intervals, and the same two contrasts are reported for strength inflation. Other comparisons, including M versus B1-R, are exploratory. Runs use greedy decoding, with a small repeated-sampling stability check.

13 Compute Plan

Table 5: Available compute (own and free resources) and estimated usage before the pilot.

Item	Plan
Own hardware	MacBook Air, Apple M5 (10-core CPU, 8-core GPU), 16 GB unified memory, ≈66 GB free storage. Used for corpus collection and parsing, BM25 indexing, annotation, evaluation scripts, and local pilots of the 8–9B model at 4-bit (≈6 GB) via MLX or llama.cpp
Free GPU (primary)	Kaggle Notebooks: 2×T4 (16 GB each), about 30 GPU-hours per week per account; used for the full experimental runs under vLLM
Free GPU (backup)	Google Colab free tier (T4, session-limited) for short debugging runs
Per query	B1 ≈ 8K input tokens; M ≈ 80K input and 9K output tokens, at most 12 tool calls
Reranker	Cross-encoder (≈0.6B parameters) re-scoring 50 passages per query; minutes on a T4
Total GPU need	At most ≈37 GPU-hours for all runs, pilots and dev iteration, within about two weeks of one Kaggle account's quota; pilots on the laptop reduce this further
Paid API	Not required. Only if no local model passes the pilot: Claude Haiku 4.5 (about USD 50, cap USD 100) *[needs TA decision]*

14 Expected Risks and Fallback Plan

Table 6: Main risks and fallbacks.

Risk	Fallback
No local model passes the pilot	Use the API model for every condition
Annotation is slower than planned	Reduce to 4 dev and 16 test queries
Low agreement on strength labels ($\kappa < 0.5$)	Score strength as direct versus hedged
Few edges at hop 3 or deeper	Pool deeper hops and report descriptively
The agent loop does not help	RQ1 is unaffected; for RQ2, report that targeted retrieval did not improve over A-NoLoop
Verifier labels nearly everything as inferred	Detected by the recall guard (H2d); tune on dev

The six-week plan runs corpus collection and pilots (week 1), annotation and baselines (weeks 2–4), frozen test runs (end of week 4), blind scoring (week 5), and analysis and writing (week 6).

Limitations: The main limitations are the small test set, a single model and time window, English-language national press only, a verifier that may share the generator's errors, and possible training-data contamination of the candidate models with 2026 news.

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